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ORIGINAL ARTICLE

The effect of low molecular weight heparin on survival in cancer patients: an updated systematic review and meta-analysis of randomized trials

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Summary. *Background:* Tumors may exploit the coagulation system to enhance the survival and dissemination of cancer cells. Some studies have suggested that heparin and low molecular weight heparin (LMWH) have antitumor effects. We reported a previous meta-analysis that suggested a modest improvement in overall survival with the use of LMWH in patients with cancer. Herein, we present the results of an updated systematic review and meta-analysis. *Objective:* To evaluate the effect of LMWH as compared with placebo or no anticoagulant on the overall survival in patients with solid cancers. *Methods:* We conducted a systematic review and meta-analysis of randomized trials evaluating the use of LMWH vs. placebo or no anticoagulant in cancer patients without venous thrombosis. A meta-analysis was conducted with a random-effects model, and data were analyzed by the use of odds ratios (ORs) and relative risks (RRs) calculated for 1-year overall mortality. *Results:* We identified 724 potentially relevant studies, nine of which met our inclusion criteria, and reported data on 1-year overall mortality. Studies were heterogeneous regarding types of cancer and interventions, and included 5987 patients, 98.4% of whom had advanced-stage disease (III and IV). There was no discernible effect on mortality with the use of LMWH (pooled OR 0.87, 95% CI 0.70–1.08; RR 0.94, 95% CI 0.86–1.04). *Conclusions:* In contrast to the previous study, these results did not show a survival benefit in cancer patients receiving LMWH. The effect of LMWH on overall survival in patients with limited-stage disease still is unknown.

Keywords: cancer; hemorrhage; low molecular weight heparin; mortality; thrombosis.

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Introduction

Patients with cancer are at significantly increased risk for venous thromboembolism (VTE) [1]. This risk varies with cancer histology, individual patient factors, treatment with chemotherapy, and the extent of disease [2]. Low molecular weight heparin (LMWH) is widely considered as a first-line treatment for VTE in patients with cancer, following results from the CLOT trial [3]. There is also evidence that LMWH thromboprophylaxis prevents VTE in some groups of patients with cancer [4,5]. Primary prophylaxis is not routinely used in all patients with cancer, but could theoretically improve survival in cancer patients by preventing VTE, particularly in patients at the highest risk. The benefit of prophylactic LMWH is balanced by the potential increased risk of bleeding and added complexity of care and expense in this group of patients.

There is also evidence that unfractionated heparin (UFH), LMWH and other anticoagulants have antiangiogenesis and antimetastatic properties [6]. Several *in vitro* and *in vivo* models of cancer suggest that cancer cells exploit the coagulation system to facilitate cell growth, angiogenesis, immune evasion, and metastasis formation [6]. Tissue factor (TF) is expressed on many tumor cells, and initiates coagulation, resulting in thrombin generation, the deposition of fibrin, and platelet activation [6]. It has been reported that tumor cells coated with platelets and fibrin may evade recognition by natural killer cells [7]. Tumor growth and angiogenesis may also be partly mediated by the TF cytoplasmic domain coupled with proteolytic activation of protease-activated receptor 2 or non-proteolytic integrin ligation [8–10]. Heparin and

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LMWH have been reported to interfere with the process of angiogenesis in tumor cells in *in vitro* experiments [11]. Also, it has been reported that tinzaparin use in mouse models of melanoma markedly reduces lung metastasis formation [12].

The clinical relevance and implications of the proposed anticancer effects of heparin and LMWH are unclear. Early studies using heparin in patients with cancer reported improved overall survival, although this has not been consistent across all studies [13–16]. We previously reported a systematic review and meta-analysis that suggested an improvement in overall survival in patients with solid malignancies receiving LMWH [17]. However, the improvement was modest, and was associated with low to moderate statistical heterogeneity. As 5 years have elapsed since our previous review, and several large trials involving the use of prophylactic LMWH in cancer patients have been recently reported, we sought to update our previous study.

Methods

Search strategy

This aim of this systematic review and meta-analysis was to evaluate the effect of prophylactic LMWH on overall survival in patients with cancer. In our meta-analysis, we included randomized controlled trials that compared LMWH with placebo or no treatment in patients with solid malignancies without active VTE. Our primary outcome was 1-year mortality, although, in contrast to our previous study we did not require this to be the primary outcome for an individual trial to be included.

The search was initially conducted in May 2013, and was updated in February 2014 with the OVID interface and including MEDLINE, EMBASE, and HealthSTAR. We searched with the terms (cancer or tumor or malignancy).tw, (heparin or low-molecular weight heparin or enoxaparin or dalteparin or reviparin or bemiparin or certoparin or tinzaparin or nadroparin or semuloparin).tw, and Random$.tw, where the suffix .tw indicates a text-word search. We also searched online databases of conference abstracts for the annual meetings of the American Society of Hematology, the American Society of Cancer and Oncology, and the ISTH. We cross-referenced from the reference lists of retrieved articles. We did not consider any limitations on the languages or dates of articles. We evaluated the retrieved references for possible inclusion on the basis of the title and abstract. If an abstract was not available, we retrieved the full article for evaluation.

Assessment of study quality and data extraction

Articles that met our inclusion criteria were retrieved for full data extraction. This was performed by one reviewer (A.N. or N.A.), and verified independently by a second reviewer (D.S.). Our inclusion criteron was as follows: randomized controlled trials comparing LMWH with placebo or no treatment in patients with solid cancer without VTE. We excluded studies if they included patients with acute VTE or hematologic malignancies. Our primary outcome was mortality at 1 year, although we extracted mortality data from retrieved studies if they were reported over a different time period. If 1-year mortality data were unavailable in a publication for a trial that met our inclusion criterion, we attempted to contact the corresponding author to obtain this data. We also extracted data on thrombotic and major bleeding events from the retrieved articles, if available. Major bleeding episodes were defined according to the ISTH criteria [18]. We assessed the quality of the studies according to the criteria proposed by Jadad *et al.* [19], and we defined allocation concealment according to the criteria proposed by Schulz and Grimes [20].

Statistical analysis

We conducted our meta-analysis with a random-effects model, as previously described by DerSimonian and Laird [21]. Our primary outcome of interest was the number of deaths at 1 year, and we calculated the pooled effect estimates as both odds ratios (ORs) and relative risks (RRs). We used a z-test to test for differences between effects, and considered a P-value of <0.05 to be statistically significant. We estimated heterogeneity by using Cochran's Q-statistic and the Higgins I^2-statistic. For Cochran's Q-statistic, we considered a P-value of < 0.1 for a chi-squared value to be indicative of heterogeneity. We defined a Higgins I^2-statistic of <25% as low heterogeneity, 25–50% as moderate heterogeneity, and > 50% as high heterogeneity [22]. Statistical analysis was performed with either Microsoft Excel or REVMAN software version 5.2.3 (The Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, Denmark). As in our previous study, we had planned to conduct a survival meta-analysis and a meta-analysis of size effects by using pooled hazard ratios (HRs), but, unfortunately, the data required to conduct these analyses were unavailable for many of the studies, and this analysis was not performed.

Results

Results of the search strategy from the systematic review

Our search strategy resulted in the identification of 724 references, and 715 of these were ultimately excluded (Fig. 1). Nine studies were included in the meta-analysis [4,5,16,23–28]. We fully evaluated 28 studies that met the criterion for inclusion in our meta-analysis, but excluded 19 of these because of unavailable data on 1-year survival outcomes [29–47]. For one of the studies, the 1-year survival data were extracted from the Kaplan–Meier survival curves [27].

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Characteristics of included studies

All of the nine studies included in the meta-analysis were randomized controlled trials with either a placebo arm or a no-treatment arm (Table 1). Seven of nine were conducted with double blinding at the outset, although one of these [26] was changed to an open-label trial midway through, owing to difficulties with recruitment. Two studies were conducted as open-label trials from the outset [16,28]. Allocation concealment was adequate in five of the studies, but in four this was unclear. Jadad's score ranged between 2 and 5 for the included studies.

A summary of the patient characteristics of included trials is shown in Table 2. The studies included 5987 patients, 3188 of whom received LMWH. Six of the studies included only patients with locally advanced or metastatic solid malignancy, and 98.4% of the included patients had stage III or IV disease [4,5,23,24,26,27]. These studies included patients with a range of different tumor histologies and primary sites, including: gastrointestinal cancer, lung cancer, gynecologic cancer, urologic cancer, prostate cancer, breast cancer, and head and neck cancers. The treatment approach for these six trials was also heterogeneous, with patients receiving a range of treatments with chemotherapy or radiotherapy or no treatment according to the discretion of the treating physician and individual patient preference. Two of the studies included only patients with previously untreated small cell lung cancer, and had a standardized approach to chemotherapy and radiation therapy for all patients [16,28]. Another study enrolled only patients with malignant glioma, and a proportion of these patients had been subjected to an initial attempt at surgical resection, and some had also received radiation therapy [25].

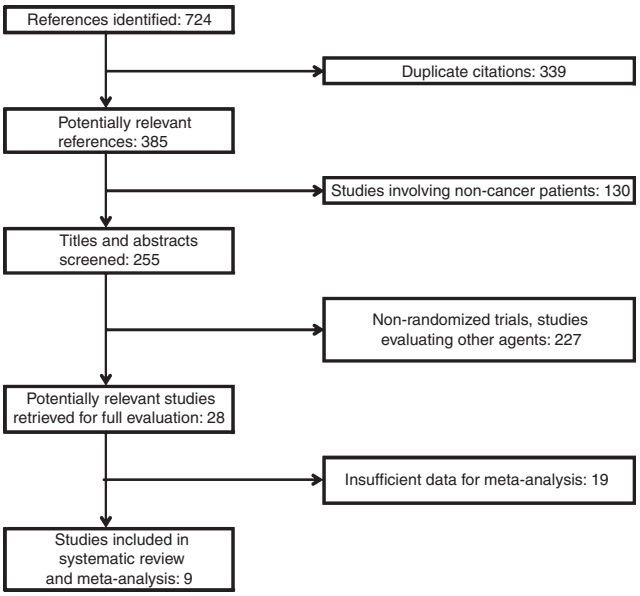

Fig. 1. Flow diagram of the systematic review.

There was wide variation in the type of LMWH used, and the onset and duration of treatment with LMWH. Four of the trials used dalteparin at a dose of 5000 units daily [16,23,25,26], three used nadroparin [4,24,27], one used semuloparin [5], and one used bemiparin [28]. The LMWH was administered daily in all trials, and the duration of treatment ranged from 6 weeks to 2 years.

Results of the meta-analysis on overall mortality

The number of participants and event rates for the nine included trials are shown in Fig. 2A,B. The pooled OR for overall 1-year mortality was 0.87 (95% confidence interval [CI] 0.70–1.08, $P = 0.21$) (Fig. 2A). The pooled RR was 0.94 (95% CI 0.86–1.04, $P = 0.24$) (Fig. 2B). For both analyses, there was statistical heterogeneity (χ^2 $P \leq 0.1$; Higgins I^2 59% and 53% for OR and RR respectively). We had planned to conduct a survival meta-analysis and a meta-analysis of size effects by using pooled HRs; unfortunately, the data required to conduct these analyses were unavailable for many of the studies, and they were not performed.

Results of the meta-analysis on VTE

We were able to extract data on VTE for the nine trials that we included in the meta-analysis on overall mortality (Fig. S1A,B). The trials all reported the total number of thrombotic events over the study period, which was different for each trial. We calculated an OR of 0.56 (95% CI 0.40–0.81) favoring the LMWH intervention. This was significant by z-test ($Z = 3.16$, $P = 0.002$). Heterogeneity testing with both Cochran's Q-statistic ($\chi^2 = 9.29$, $P = 0.32$) and the Higgins I^2-statistic (14%) suggested low heterogeneity. Calculating the RR of VTE, we estimated the effect at an RR of 0.59 (95% CI 0.42–0.83), which was also significant ($z = 3.04$, $P = 0.002$).

Results of the meta-analysis on major bleeding

We extracted data on major bleeding events from the nine trials that we included in the meta-analysis on overall mortality (Fig. S2A,B). We calculated an OR of 1.22 favoring no LMWH intervention (95% CI 0.74–2.02), but this was not significant ($P = 0.39$), with low heterogeneity. We calculated a pooled RR of 1.21 (95% CI 0.75–1.96), which was also not significant ($P = 0.44$).

Sensitivity and influence analyses

We estimated overall mortality for six of the studies when excluding the three studies that enrolled patients with only a single type of primary cancer [16,25,28]. This did not change the results of our meta-analysis, and the pooled OR was 0.93 (95% CI 0.81–1.07, $P = 0.33$). We

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Table 1 Characteristics of included studies

	Agnelli *et al.* [4]	Agnelli *et al.* [5]	Altinbas *et al.* [16]	Kakkar *et al.* [23]	Klerk *et al.* [24]	Lecumberri *et al.* [28]	Perry *et al.* [25]	Sideras *et al.* [26]	van Doormaal *et al.* [27]
Design	Double-blind RCT	Double-blind RCT	Open-label RCT	Double-blind RCT	Double-blind RCT	Open-label RCT	Double-blind RCT	Double-blind RCT l; open-label RCT	Open-label RCT
No. of participants (controls)	1150 (381)	3212 (1604)	82 (42)	374 (184)	302 (154)	38 (18)	186 (87)	Double-blind phase: 50 (26) Open-label phase: 88 (44)	503 (259)
Jadad score	5	4	2	5	5	3	3	2	3
Allocation concealment	Adequate	Unclear	Unclear	Adequate	Adequate	Unclear	Adequate	Adequate	Unclear
Inclusion criteria	Metastatic or locally advanced solid tumor	Metastatic or locally advanced solid tumor	Small cell lung cancer, previously untreated	Metastatic or locally advanced solid tumor	Advanced incurable solid tumor	Limited-stage small cell lung cancer	Malignant glioma	Advanced incurable solid tumor	Non-small cell lung cancer (stage IIIB), hormone-refractory prostate cancer, or advanced pancreatic cancer
Interventions	Nadroparin 3800 U once daily given for duration of CT up to a maximum of 4 months	Semuloparin 20 mg once daily until change in CT regimen	Dalteparin 5000 U once daily for 18 weeks	Dalteparin 5000 U once daily for 1 year or until death	Weight-based nadroparin (3800–7600 U) twice daily for 2 weeks and then once daily for 4 weeks	Bemiparin 3500 units once daily for 26 weeks	Dalteparin 5000 U once daily for 6 months	Dalteparin 5000 U once daily for 2 years or until death	Weight-based nadroparin for 2 weeks (7600–15 200 U TDD) and then half-therapeutic doses for 4 weeks (3800–7600 U TDD)
Concomitant therapies	Usual CT	Usual CT	Patients received six cycles of cyclophosphamide, epirubicin, and vincristine. Patients with limited disease received RT if they responded after CT	Usual CT, RT, or surgery	Usual CT, RT, and hormone therapy, and other therapeutic modalities	Concurrent platinum-based CT and RT	Some patients had surgical resection and/or adjuvant RT	Standard CT or RT: 29 dalteparin and 31 control patients were not receiving any treatment at inclusion	Usual CT or RT

CT, chemotherapy; RCT, randomized control trial; RT, radiotherapy; TDD, total daily dose.

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Table 2 Characteristics of patients included in randomized trials evaluating the impact of low molecular weight heparin on cancer survival

	Agnelli *et al.* (2009) [4]		Agnelli *et al.* (2012) [5]		Perry *et al.* [25]		Van Doormal *et al.* [27]		Altinbas *et al.* [16]		Kakkar *et al* [23]		Klerk *et al* [24]		Sideras *et al* [26]		Lecumberri *et al* [28]	
	LMWH group	Control group	LMWH group	Control group	LMWH group	Control group	LMWH group	Control group	LMWH group	Control group	LMWH group	Control group	LMWH group	Control group	LMWH group	Control group	LMWH group	Control group
Median age at entry (years)	62.1	63.7	59.8	59.4	57	55	65	65	57.5	58	62	60.9	63	64	64.5 (68.5)*	63.5 (70.3)*	61.1	64.5
Male gender (%)	48.4	48	60.6	59.6	62	57	81	80	78.6	83.3	40.5	45.7	52	53	58.8	60	85	89
Limited Disease (stage I-II) (%)	0	0	0	0	Glioma	Glioma	0	0	54.8	59.5	3.2	2.1	0	0	0	0	100	100
Median overall survival (months)	NR	NR	NR	NR	NR	NR	13.1	11.9	13	8	10.8	9.1	8	6.6	7.3	10.5	37.2	11.3
Performance status at inclusion (%)																		
Scale used†	NR		ECOG		Karnofsky		Karnofsky		ECOG		NR		WHO		ECOG		ECOG	
0/≤1	NR	NR	40.2	40.6	NA	NA	NA	NA	7.2	12	NR	NR	86.5	83.1	32.4	31.4	35	22.2
1/2/≤80	NR	NR	51.1	51.7	45	38	36.1	37.8	61.9	57.1	NR	NR	10.1	11.7	54.4	50	65	78.8
>2/≥3/>80	NR	NR	8.7	7.1	55	62	63.9	62.2	30.9	30.9	NR	NR	8	5.2	13.2	18.6	0	0
Type of cancer (%)																		
Breast	14.3	14.4	NR	NR	0	0	0	0	0	0	21.1	14.1	20.9	12.3	11.8	10	0	0
Colorectal	27.8	28.2	28.9	28.7	0	0	0	0	0	0	18.4	19	12.2	22.7	25.0	27.1	0	0
Ovarian	12.5	12.3	11.9	11.7	0	0	0	0	0	0	17.4	15.2	2	3.9	0	0	0	0
Pancreatic	4.7	4.5	7.8	8	0	0	25.8	27.8	0	0	9.5	13	5.4	6.5	0	0	0	0
Lung (Small Cell)	25.9	21	36.8	36.7	0	0	33.2	34	100	100	NR	NR	12.2	10.4	8.8	5.7	100	100
Other	14.8	19.6	14.6	14.9	100	100	41	38.2	0	0	34	39	47.3	44.2	54.4	57.2	0	0
Concomitant therapies (%)																		
Chemotherapy	100	100	100	100	0	0	NR	NR	100	100	33.7	31	25	34.4	51.5	54.3	100	100
Radiotherapy	0	0	0	0	78	90	NR	NR	0	0	8.4	7.6	32.4	18.2	8.8	5.7	90	83.3
Other	0	0	0	0	22	10	NR	NR	0	0	1.6	1.1	16.9	16.2	NR	NR	NR	NR

NR, Not Reported; NA Not applicable; WHO, World Health Organization; ECOG, Eastern Cooperative Oncology Group; LMWH, Low Molecular Weight Heparin.
*The figures in brackets indicate the data for the open-label patients.
†The order of the cutoff values for the scores shown below is: ECOG/WHO/Karnofsky.

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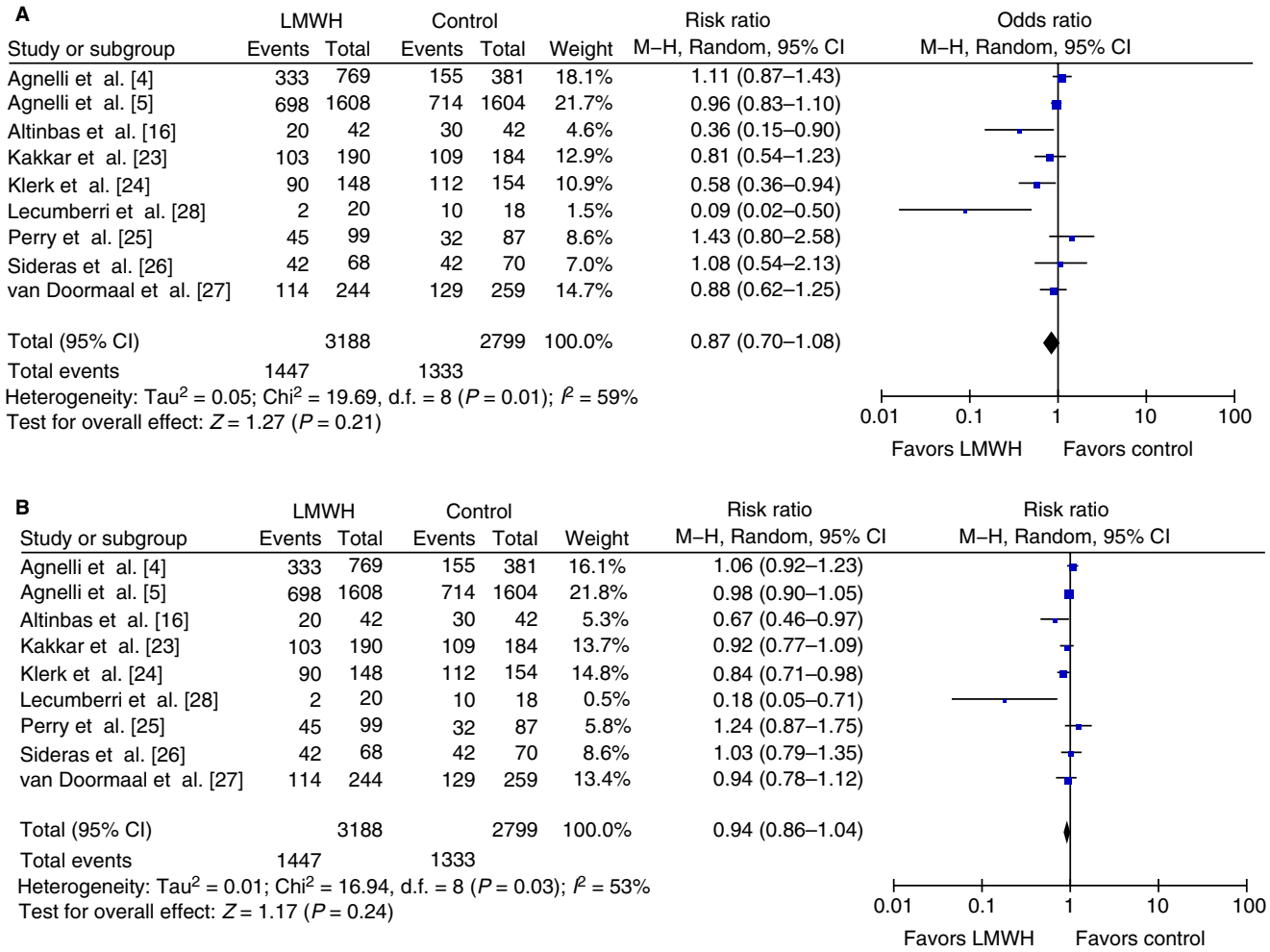

Fig. 2. Forest plots for overall 1-year mortality in cancer patients randomized to low molecular weight heparin (LMWH) vs. placebo/no intervention. (A) Pooled odds ratio of death obtained using a random-effects model. (B) Pooled risk ratio of death obtained using a random-effects model. CI, confidence interval; d.f., degrees of freedom; M-H, Mantel-Haenszel.

also estimated the pooled OR after excluding five of the studies with lower estimated methodological quality (Jadad score of < 4) [16,25–28], and we found an OR for the four remaining studies of 0.91 (95% CI 0.75–1.12, $P = 0.37$). We also performed an influence analysis, excluding individual studies from the meta-analysis, and this suggested that no single study dominated the combined results. The results of this are shown in Table 3.

Table 3 Influence analysis for effect of individual studies

Study excluded	Pooled OR (95% CI)	*Z*-value (*P*-value)	Pooled RR (CI)	*Z*-value (*P*-value)
Agnelli *et al.* [4]	0.81 (0.63–1.06)	1.55 (0.12)	0.92 (0.83–1.03)	1.48 (0.14)
Agnelli *et al.* [5]	0.82 (0.61–1.11)	1.30 (0.19)	0.93 (0.82–1.06)	1.08 (0.28)
Altinbas *et al.* [16]	0.91 (0.75–1.12)	0.88 (0.38)	0.96 (0.88–1.05)	0.83 (0.41)
Kakkar *et al.* [23]	0.87 (0.68–1.11)	1.11 (0.27)	0.95 (0.85–1.06)	0.96 (0.33)
Klerk *et al.* [24]	0.92 (0.74–1.14)	0.76 (0.45)	0.96 (0.87–1.07)	0.69 (0.49)
Lecumberri *et al.* [28]	0.91 (0.77–1.09)	1.00 (0.32)	0.95 (0.88–1.03)	1.16 (0.25)
Perry *et al.* [25]	0.83 (0.66–1.04)	1.62 (0.11)	0.93 (0.84–1.03)	1.48 (0.14)
Sideras *et al.* [26]	0.85 (0.67–1.07)	1.36 (0.17)	0.93 (0.84–1.04)	1.25 (0.21)
van Doormaal *et al.* [27]	0.86 (0.66–1.10)	1.20 (0.23)	0.94 (0.84–1.05)	1.03 (0.30)

CI, confidence interval; OR, odds ratio; RR, relative risk. Individual studies were excluded systematically from the meta-analysis to evaluate the influence of each study. Pooled ORs and RRs were calculated for the remaining studies.

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Discussion

In contrast to our previous study, the results of our updated meta-analysis suggest that the use of prophylactic LMWH does not have a discernible effect on overall survival in patients with solid malignancy without VTE. Our previous meta-analysis found a modest increase in overall survival, with a calculated OR of 0.70 (95% CI 0.49–1.00, $P = 0.05$), an RR of 0.87 (95% CI 0.77–0.99, $P = 0.04$), and an HR of 0.83 (95% CI 0.7–0.99, $P = 0.03$). As anticipated, LMWH use did appear to reduce thrombotic events when the data from the included trials were pooled, and there was a trend towards increased major bleeding with LMWH, although this was not significant. Our findings are consistent with a previous Cochrane review and meta-analysis, which did not report improved survival associated with LMWH in patients with cancer [48,49]. Both of these studies reported a significant decrease in the rate of VTE associated with the use of LMWH, but this was balanced by a non-significant trend towards increased bleeding with LMWH. Our meta-analysis included more trials, an extensive statistical analysis, and additional outcomes of interest, but the conclusions from these studies validate our current findings.

As compared with our previous report, the current meta-analysis included five additional studies [4,5,25,27,28]. Three of the newly included studies were large in comparison with the previous studies; they enrolled 3212, 1150 and 503 patients, respectively, and were assigned a higher weight in the meta-analysis [4,5,11]. In these three studies, there was no statistically significant difference or trend in difference in overall survival with LMWH use. A more recent, smaller study included only patients with malignant glioma, and reported a trend towards increased mortality in the LMWH group, although this did not reach statistical significance within the individual study [25]. Another, more recent, trial included patients with limited-stage small cell lung cancer, and reported a significant overall survival advantage in the group assigned to LMWH, although the trial included only 38 patients [28].

Only three of the individual studies reported a statistically significant improvement in overall survival in patients randomized to LMWH [16,24,28]. One of these studies included patients with different types of advanced, non-curable solid malignancies, and there was a heterogeneous approach to anticancer therapy [24]. The other two studies involved patients with small cell lung cancer, and reported a standardized approach in the treatment of the patients [16,28]. Interestingly, a previous trial also reported improved survival with the use of UFH in patients with small cell lung cancer [14]. This trial randomized 277 patients with small cell lung cancer to 5 weeks of subcutaneous UFH vs. no heparin for 5 weeks, starting with their first cycle of chemotherapy. The authors of this study also reported improved rates of complete remission in the heparin group (37% vs. 23%, $P = 0.004$), and longer median overall survival (317 days vs. 261 days, $P = 0.01$). The control group in this trial contained slightly more male patients and slightly more patients with extensive disease, perhaps influencing the results. The study by Altinbas *et al.* also reported higher overall response rates to chemotherapy in patients treated with LMWH (69.2% vs. 42.5%, $P = 0.07$), whereas no significant difference in response was reported in the study by Lecumberri *et al.* [16,28]. The results of the FRAGMATIC trial investigating the addition of 24 weeks of prophylactic dalteparin to standard therapy in small cell and non-small cell lung cancer patients may help to provide more information on the impact of LMWH in this group of patients [37].

Five of the trials included in the meta-analysis enrolled cancer patients with a wide variety of primary sites and histologies. None of these individual trials demonstrated a significant benefit for overall mortality with the use of LMWH, although two of these did, perhaps, show a trend towards improved overall survival [23,24]. All of these trials involved patients with either metastatic or locally advanced disease, and the majority of these patients would probably be considered to be incurable and have a relatively unfavorable overall prognosis and shorter life-expectancy. When the results of these trials are pooled, there is no evidence that administering LMWH heparin to all patients with advanced malignancy will improve survival, although our meta-analysis suggests a reduction in thrombotic events with this intervention.

The discrepancies among our current and previous results might be explained by the fact that the current study included far more patients in total than the previous one. Furthermore, the pooled HR estimates in the previous studies were associated with significant statistical heterogeneity (55.4% by I^2-test). The influence analysis in our previous study suggested that, after exclusion of individual studies, pooled effect size estimates might have been non-significant. In the present study, the sample size resulted in homogeneous results, suggesting that, indeed, addition of more patients to the meta-analysis showed no effect of LMWH on survival. As in our previous study, data available from individual studies lacked uniformity, and information was missing. Nineteen of the studies that we retrieved for data extraction did not report 1-year mortality, and we were unable to include these in our meta-analysis [29–47]. We attempted to contact the corresponding authors of these studies to obtain 1-year mortality data, but either these were unavailable or we received no response. We are uncertain of the effect that these studies may have had on our results had these data been available. As only one of these studies reported improved overall survival with LMWH, we can only speculate that the overall results would have been unchanged. We were not able to include data from the TOPIC 1 and TOPIC 2

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trials, two of the larger and more recent trials in this area [30]. These trials tested the effect of prophylactic certoparin on VTE occurrence in patients with metastatic breast cancer and advanced non-small cell lung cancer, respectively. Neither of these trials reported a mortality difference at 6 months, although longer-term mortality data were unavailable. Additionally, we cannot determine the effect of LMWH on survival among different cancer types, as information for individual groups was not available.

Several limitations of our study should be noted. We cannot rule out a differential antitumor effect between different LMWH preparations. A previous study reported that very low molecular weight heparin preparations are not as effective as higher molecular weight ones, and, indeed, preparations with a molecular mass of < 3 kDa did not have an antitumor effect *in vitro* [50]. This is particularly relevant, given that the SAVE-ONCO study used semuloparin, an ultra-low weight agent with an average molecular mass of 2–3 kDa. It is also unclear whether the use of higher doses of LMWH might provide greater antitumor activity or possibly improve overall survival. One of the studies included in the meta-analysis used a short treatment period with weight-based nadroparin, and reported a modest trend towards improved overall survival with this regimen [27]. Another study identified in our systematic review also reported a modest, non-significant improvement associated with the use of weight-based dalteparin in patients with pancreatic cancer [31]. Interestingly, both trials reported similar rates of major bleeding in patients receiving weight-based LMWH and in controls, suggesting that future studies testing a higher dose of LMWH in patients with limited-stage disease might be feasible from a safety perspective [27,31]. Finally, it is entirely possible that LMWH has a beneficial effect across specific subtypes of cancer or in patients with limited-stage disease. This was suggested by the fact that the two studies including only patients with small cell lung cancer, most of whom were at an early stage, showed a non-significant risk reduction (RR 0.40, 95% CI 0.11–1.53, $P = 0.18$) and a reduction in the OR (OR 0.22, 95% CI 0.06–0.83, $P = 0.03$) for 1-year mortality associated with LMWH [16,28]. These results, however, came from a *post hoc* analysis, were associated with moderate to high statistical heterogeneity, and were based on a rather small number of patients ($n = 122$); therefore, they should be regarded only as hypothesis-generating.

Future trials should attempt to focus on a single cancer type with a defined treatment strategy for the enrolled patient group, to allow the results of these types of trial to be more easily interpretable. Studies investigating the effect of LMWH on survival in cancer patients should include patients with limited-stage disease, as this group has not been represented in the published trials. Currently, two ongoing studies are attempting to partially answer this question: the PERIOP-01 study (NCT01455831), which is investigating the impact of longer-term use of prophylactic tinzaparin after surgical resection of colon cancer with curative intent, and the TILT study (NCT00475098), which is assessing the impact of tinzaparin on survival in patients with localized non-small cell lung cancer [34,51]. Finally, we emphasize again the importance of providing all pertinent information in order to facilitate future comprehensive evaluations of the literature.

In summary, in contrast to our previous study, this updated meta-analysis does not suggest that LMWH improves survival in patients with advanced solid tumors. Many questions remain to be answered, including the role of LMWH for individual tumor types, the effects of the individual LMWH type, dose, and duration, and, most importantly, the effect on patients with limited-stage disease.

Addendum

D. Sanford and A. Lazo-Langner designed the study, performed the statistical analysis for the meta-analysis, and wrote the manuscript. D. Sanford, N. Alizadeh, and A. Naidu performed the systematic review. A. Naidu and D. Sanford extracted data from the retrieved articles for the meta-analysis. All authors contributed to editing and revising the manuscript.

Disclosure of Conflict of Interests

A. Lazo-Langner has received honoraria from Pfizer, Leo Pharma, and Boehringer Ingelheim, research support from Alexion, and has participated in studies sponsored by Pfizer, Leo Pharma, Bayer, Daiichi-Sankyo, Novartis, and Celgene. None of these entities were involved in any aspect of this study. All other authors state that they have no conflict of interest.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Fig. S1. Forest plots for venous thromboembolism in cancer patients randomized to low molecular weight heparin (LMWH) vs. placebo/no intervention. Pooled odds ratio (A) and risk ratio (B) were obtained with a random-effects model. CI, confidence interval; d.f., degrees of freedom; M-H, Mantel-Haenszel.

Fig. S2. Forest plots for major bleeding in cancer patients randomized to low molecular weight heparin (LMWH) vs. placebo/no intervention. Pooled odds ratio (A) and risk ratio (B) were obtained with a random-effects model. CI, confidence interval; d.f., degrees of freedom; M-H, Mantel-Haenszel.

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